scheduling.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[2]

```
import random

n = int(input("Enter number of processes: "))

burst = []

for i in range(n):
    time = int(input(f"Enter burst time for P{i+1}: "))
    burst.append(time)


def waiting_time(order):
    time = 0
    total = 0

    for i in order:
        total += time
        time += burst[i]

    return total


particles = []

for i in range(10):
    order = list(range(n))
    random.shuffle(order)
    particles.append(order)


best_order = particles[0]
best_time = waiting_time(best_order)

for order in particles:
    time = waiting_time(order)

    if time < best_time:
        best_time = time
        best_order = order
```

colab.research.google.com/drive/19lkmdYOc4g2Aq5HXxqwzeQ3Ww5v7DDZZ#scrollTo=EmDx_PGl6wZT

scheduling.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[2]

```
        if time < best_time:
            best_time = time
            best_order = order


    print("\nBest CPU Schedule:")

    for i in best_order:
        print("P" + str(i + 1), end=" ")

    print("\nTotal Waiting Time:", best_time)
    print("Average Waiting Time:", best_time / n)
```

```
Enter number of processes: 4
Enter burst time for P1: 4
Enter burst time for P2: 5
Enter burst time for P3: 2
Enter burst time for P4: 3

Best CPU Schedule:
P3 P1 P4 P2
Total Waiting Time: 17
Average Waiting Time: 4.25
```